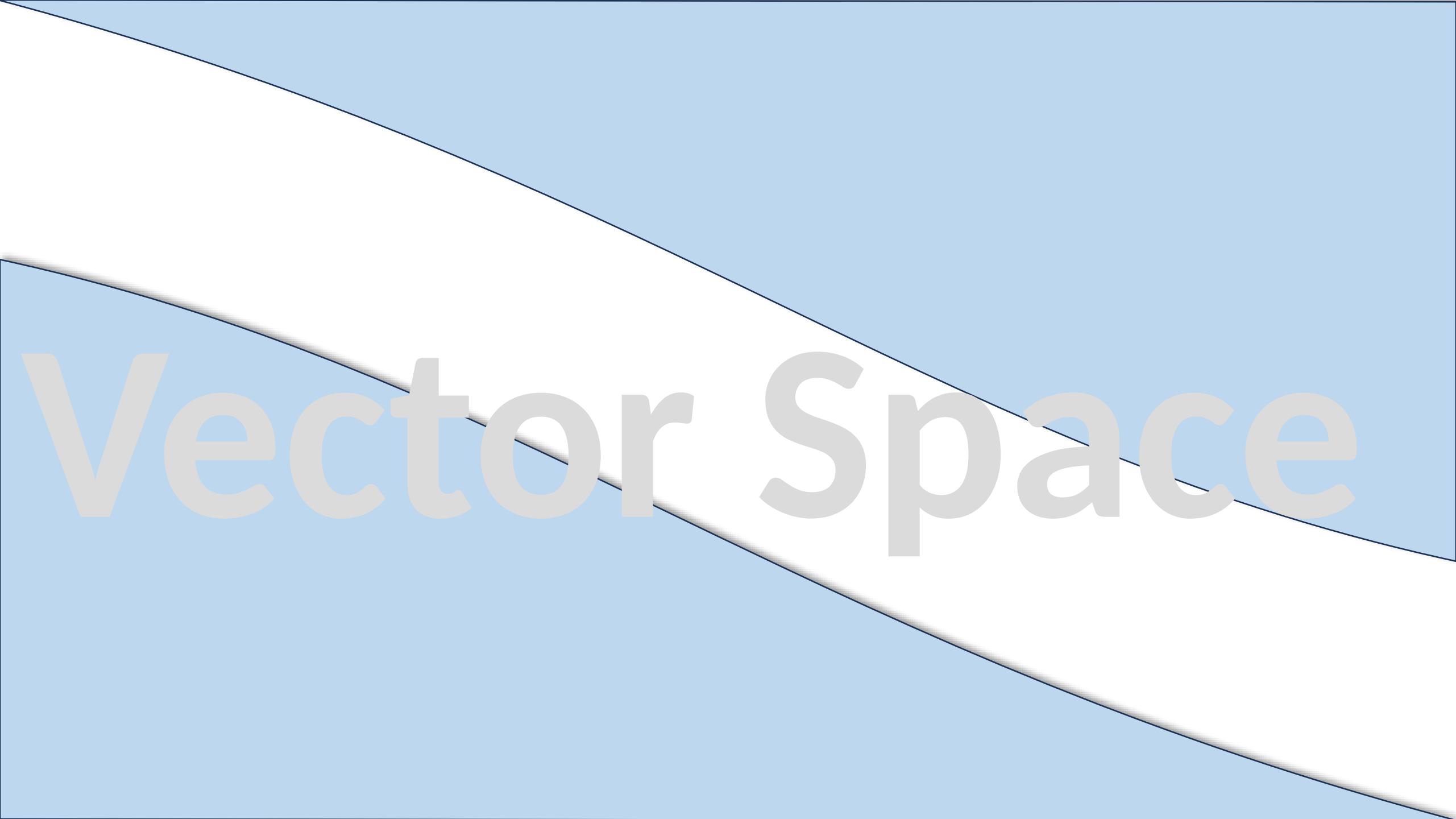


Vector Space

The background of the image consists of two broad, diagonal stripes. One stripe is a light blue color, and the other is white. They intersect to form a large 'X' shape across the entire frame. The stripes are separated by thin, dark blue lines.

Introduction to Vector Space

“Dear students, imagine you’re designing an AI system for a robot that navigates a smart city.”

This robot doesn’t just move randomly. It processes **sensor data**, **location info**, **object recognition vectors**, and **user commands**—all as **vectors**. And these vectors live in a special world: a **vector space**.

Just like a video game world has rules—gravity, collision detection, physics—the world of data and AI also has rules.

Those rules define how your information behaves, combines, and scales.

Introduction (Let's go deeper)

When your robot sees two paths and blends the information to make a decision?

👉 That's **vector addition**.

When you adjust the importance (weight) of a feature in a machine learning model?

👉 That's **scalar multiplication**.

And when your data follows consistent, predictable patterns under these operations?

👉 You've built a **vector space**.

Just like physical space defines the rules for moving real-world objects, a

vector space defines the rules for manipulating digital objects

whether it's text, audio, image, or behavior in a simulation.

What is a Vector Space?

A **vector space** (or linear space) is a set of **vectors**, along with two operations—**vector addition** and **scalar multiplication**—that satisfy certain axioms (rules).

Formally, a set V is a **vector space** over a field F (like $\mathbb{R}$ or $\mathbb{C}$) if for any vectors $\vec{u}, \vec{v}$ and $\vec{w} \in V$ and scalars $k, m \in F$, the following properties hold:

w. r. t. Addition

1. $\vec{u} + \vec{v} \in V$ **closure**
2. $(\vec{u} + \vec{v}) = (\vec{v} + \vec{u})$ **Commutativity**
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ **Associativity**
4. $\vec{0} \in V$ such that $\vec{0} + \vec{u} = \vec{u} + \vec{0} = \vec{u}$ **Identity**
5. for $\vec{u} \in V$, there exist $-\vec{u} \in V$,
such that $\vec{u} + (-\vec{u}) = \vec{0}$ **Inverse**

w. r. t. Multiplication

6. $k * \vec{u} \in V$ **closure**
7. $k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$
8. $(k + m) \vec{u} = k \vec{u} + m \vec{u}$
9. $k(m \vec{u}) = (k * m) \vec{u}$
10. $1 * \vec{u} = \vec{u}$ **Identity**

Example: Designing Movement in a 2D Game

“Imagine you’re coding movement for a 2D game character. The character can move along a 2D plane — left-right and up-down. Each position is represented as vector: $\vec{v} = (x, y)$ ”

Where x :Horizontal Position, y : Vertical Position

Vector Addition (Moving diagonally)

$$\vec{v}_1 = (2,3), \vec{v}_2 = (-1,1)$$

$$\vec{v}_1 + \vec{v}_2 = (2 - 1, 3 + 1) = (1,4)$$

Still a 2D position (still in the same space).

Scalar Multiplication (Speed boost):

Let’s apply a speed factor of 3:

$$3 * \vec{v} = 3(2,3) = (6,9)$$

Still a 2D position. Still in the same space.

This system of 2D movement vectors is:

- Closed under addition
- Closed under scalar multiplication
- Has a zero vector: $(0,0)$
- Has additive inverses

This is what we call a *vector space*.

Example: Designing a Feature Vector Space for a Face Recognition App

You've just been hired by a startup that builds **AI-powered face recognition**. Your first task?

Build the system's foundation: a **feature space** where facial data will live and be processed. Your boss tells you:

'Make sure this space behaves nicely — so we can combine, scale, and analyze facial data reliably.'

Mathematical terms: You must prove, feature vectors live in a **vector space** over $\mathbb{R}$

Each face is represented by a **vector of 3 features**: $\vec{f} = (\text{eye width}, \text{nose height}, \text{chin angle})$

Let $V = \mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$,

Your Job: Verify that V is a vector space under regular addition and scalar multiplication i.e.

$$\vec{u} + \vec{v} = (u_1, u_2, u_3) + (v_1, v_2, v_3) = (u_1 + v_1, u_2 + v_2, u_3 + v_3)$$

$$k\vec{u} = k(u_1, u_2, u_3) = (ku_1, ku_2, ku_3)$$

Solution...

Axiom	What to check (Face Recognition Terms)	Proof idea
1. Closure under Addition	Combine two faces' features — does the result still look like a valid feature vector?	Yes, sum of reals is real $\rightarrow$ result stays in $\mathbb{R}^3$
2. Commutativity	Order of combining two face feature vectors doesn't matter?	$(\vec{u} + \vec{v}) = (\vec{v} + \vec{u})$
3. Associativity	Add 3 vectors — is grouping irrelevant?	$(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
4. Existence of Zero Vector	Is there a “neutral” face with 0 for all features?	$\vec{0} = (0,0,0) \in \mathbb{R}^3$

Solution...cont

Axiom	What to check (Face Recognition Terms)	Proof idea
5. Existence of Additive Inverse	Can we subtract a face from itself to get neutral data?	$\vec{u} + (-\vec{u}) = \vec{0}$
6. Closure under Multiplication	Apply a brightness/scaling factor to a face — is it still valid data?	Yes, scaling reals stays in $\mathbb{R}^3$
7. Scalar Distributivity over Vector Addition	Apply filter $\mathbf{a}$ to $\vec{u} + \vec{v}$, same as applying to each then adding?	$a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$
8. Scalar Distributivity over Scalars	Applying $a + b$ scaling vector = applying separately then summing	$(a + b) \vec{u} = a \vec{u} + b \vec{u}$

Solution...cont

Axiom	What to check (Face Recognition Terms)	Proof idea
9. Associativity of Scalar Multiplication	Scaling by ab = first scale by b , then by a ?	$a(b \vec{u}) = (a * b) \vec{u}$
10. Identity Element of Scalar Multiplication.	Apply factor 1 -- does the vector stay the same?	$1. \vec{u} = \vec{u}$

Identity: The **neutral vector** would be: $\vec{0} = (0,0,0)$ If you add this to any facial feature vector $\vec{f}$, the features remain unchanged so it acts as the **neutral element** for addition.

Inverse: Imagine if you extract features from a face — eye width, nose height, and chin angle — and you **reverse** all of them exactly. The result is like **undoing the original face features**. When you combine both, it's like canceling out the original data — leaving you with a **neutral/no-effect vector**: the zero vector.

Example: Corrupted Data in a 2D Sensor System

You're building a 2D tracking system for a drone's position. Normally, each position is given by:

$$\vec{V} = \{(x, y) | x, y \in \mathbb{R}\}$$

However, due to a sensor failure, **the Y-axis (vertical position) is corrupted during scaling operations.**

The system can **add** positions normally, but when you try to **scale** a position by a factor (e.g., zooming out), the Y value is **lost** — it becomes 0.

Defined Operations for vectors $\vec{u} = (x_1, y_1)$, $\vec{v} = (x_2, y_2)$

Addition (Normal): $\vec{u} + \vec{v} = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$

Scalar Multiplication (Corrupted): $k\vec{u} = k * (x_1, y_1) = (k * x_1, 0)$

Check whether $\vec{V}$ is a Vector Space or not.

Solution: All properties of addition are satisfied. (Check it by yourself)

Let's check the properties of scalar multiplication.

6. Let $\vec{u} = (x_1, y_1)$ in V , then $k\vec{u} = k(x_1, y_1) = (kx_1, 0) \in V$.

7. $k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$

L.H.S. $k(\vec{u} + \vec{v}) = k((x_1, y_1) + (x_2, y_2))$
 $= k(x_1 + x_2, y_1 + y_2)$
 $= (kx_1 + kx_2, 0)$

R.H.S.

$$k * \vec{u} + k * \vec{v} = k(x_1, y_1) + k(x_2, y_2)$$
$$= (kx_1, 0) + (kx_2, 0)$$
$$= (kx_1 + kx_2, 0) \text{ so verified}$$

8. $(k + m) \vec{u} = k \vec{u} + m \vec{u}$

L.H.S. $(k + m)\vec{u} = (k + m)(x_1, y_1)$
 $= ((k + m)x_1, 0)$
 $= (kx_1 + mx_1, 0)$

R.H.S.

$$k * \vec{u} + m * \vec{u} = k(x_1, y_1) + m(x_1, y_1)$$
$$= (kx_1, 0) + (mx_1, 0)$$
$$= (kx_1 + mx_1, 0) \text{ so verified}$$

Solution: cont...

$$9. k(m \vec{u}) = (k * m) \vec{u}$$

$$\begin{aligned}\text{L.H.S. } k(m * \vec{u}) &= k(m(x_1, y_1)) \\ &= k(m * x_1, 0) \\ &= (km * x_1, 0)\end{aligned}$$

R.H.S.

$$\begin{aligned}(k * m)\vec{u} &= (k * m)(x_1, y_1) \\ &= (km * x_1, 0) \text{ so verified}\end{aligned}$$

$$10. 1 * \vec{u} = \vec{u}$$

$$\begin{aligned}\text{L.H.S. } 1 * \vec{u} &= 1 * (x_1, y_1) \\ &= (x_1, 0) \\ &\neq \vec{u}\end{aligned}$$

So, not verified

As 10th property is failed Hence $\vec{V}$ is not a Vector Space.

Real-World Insight

This corrupted data system is like trying to scale an image where one axis disappears — the operation is flawed. Just like faulty software, this set fails to meet the basic rules required to build consistent mathematical models.

Please see the MS Word file for further practice problems

THANKS